

Basil T Ebinesar

Greater Chicago Area | ebasiltamilselvan@gmail.com | +1 (219)-368-9239 | basiltamilselvan.netlify.app
linkedin.com/in/basil-tamil-selvan-226b191bb | github.com/basilll007

SUMMARY

Graduate researcher at the intersection of RNA biology, computational genomics, and biomedical AI. 1st place, BioNLP 2026 Shared Task (MMR); 2nd place, OmniRNA Discovery Challenge, ECML PKDD 2026; spotlight presentation, PharML Workshop, ECML PKDD 2026. Developing AI systems that connect sequence-level molecular biology to clinical outcomes spanning RNA inverse design, epitranscriptomic survival prediction, and multimodal biomedical retrieval. Seeking a PhD to build rigorous computational methods that are shaped by biological structure rather than merely fitted to biological data.

EDUCATION

Purdue University Northwest

Expected May 2027

M.S. in Applied Artificial Intelligence

- **Graduate Research Assistant:** Focused on Biomedical NLP and AI-driven scientific knowledge extraction under Dr. Keyuan Jiang.
- **Master's Thesis:** Spatial transcriptomics analysis of cell–cell communication (CCC) in tissue — evaluating whether ligand–receptor significance scores hold up under spatially-explicit null models versus standard cluster-mean scoring, using 10x Xenium breast tissue data.
- **Ongoing Work:** *Spatial Adjacency Reranks Ligand–Receptor Communication Significance* — a controlled null-model comparison showing that compositional CCC scoring manufactures significance for spatially unsupported interactions (e.g., CD274–PDCD1), while foundation-model (scGPT) and graph-learning baselines fail to recover the spatial-contact signal. (*Student abstract, in progress.*)
- **Research Focus:** Developing *mApIt*, an AI-powered platform for extracting, structuring, and analyzing biomedical literature (PubMed, ClinicalTrials.gov) to identify research trends and knowledge gaps.
- **Selected Coursework:** Generative AI, Practical Deep Learning.

Anna University (Kamaraj College of Engineering and Technology)

Aug 2020 – Apr 2024

B.E. in Computer Science and Engineering

- **Gold Medalist (Valedictorian):** Awarded for securing **Rank 1** out of the entire graduating batch (1000 students).
- **Distinction:** First Class with Distinction (Overall: 87%).
- **Relevant Math & Science:** Engineering Mathematics I & II (Calculus, Linear Algebra), Probability & Statistics, Engineering Physics.
- **Undergraduate Research:** *3D CNN for Brain Tumor Classification* — Developed a volumetric segmentation model for MRI scans; integrated YOLOv5 for real-time localization.
- **Awards:** Smart India Hackathon (SIH) 2022 **National Finalist**; Best Outgoing Student (Dept. of CSE).

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Sep 2025 – Present

Purdue University Northwest | Advisor: Dr. Keyuan Jiang

- **Research Focus:** Developing **AI-driven systems for biomedical knowledge extraction and analysis**, leveraging large-scale scientific literature (PubMed, ClinicalTrials.gov), including work on **mApIt** and **Author Name Disambiguation (AND)** using hybrid (rule-based + LLM) approaches.

Research Trainee

[Start Date] – Present

Mayo Clinic — Collaborative Project with Purdue University Northwest

- Selected as a **Research Trainee** on a joint Mayo Clinic–PNW initiative, working with **Dr. Bryan Farford** and **Dr. Amir MajidiRad** (Mayo Clinic) on two clinical decision-support projects.
- **ATM Mayo:** Built a **Flutter/Firebase** healthcare application connecting patients, nurses, and medication-dispensing machines, including Google Maps integration and a planned **FastAPI** LLM microservice.
- **RPM (Remote Patient Monitoring for CHF):** Co-authored "A Decision-Support Remote Patient Monitoring Framework for Congestive Heart Failure to Enable Earlier Clinical Detection," a prototype platform with role-based (patient/nurse/admin) dashboards and rule-based clinical alert logic (weight change, SpO₂ thresholds), built with **Python + FastAPI, PostgreSQL**, JWT auth, and a **React Native (Expo)** frontend.

AI/ML Intern

May 2026 – Present

Rarelife Solutions — Remote

- Building **production AI/ML platforms** for pharmaceutical clients, including **Insight**, a competitive intelligence tool that ingests and deduplicates competitor drug data from **ClinicalTrials.gov** and **OpenFDA**.
- Developed the **mApIt** pipeline for generating publication-landscape executive summary PDFs for pharma clients.

AI Researcher (Computer Vision)

Nov 2024 – Jun 2025

H1 Enterprise — Remote

- Conducted R&D using **computer vision techniques** for agricultural applications, focusing on plant health monitoring and disease detection from leaf images.
- Developed deep learning models for identifying plant leaf diseases and analyzing visual patterns to support early-stage crop diagnostics.
- Explored scalable approaches for real-world deployment, including optimization of models for efficient inference in resource-constrained environments.

PUBLICATIONS

- Sai Suresh Macharla Vasu, **Basil Ebinesar**, Hemashruthi Durairaj, Charansai Maddineni, Keyuan Jiang. *Detection Is Not Resolution: Contextual Trust in Long-Term Memory*. *NeurIPS 2026 Workshop on Women in Machine Learning (WiML)*, 2026. (Accepted, poster presentation)
- **Basil Ebinesar**, Keyuan Jiang, Charansai Maddineni, Ashok Raja. *Pride-Boiler at MedGenVidQA 2026: LLM-Augmented BM25 Retrieval with Corrective Self-Verification for Biomedical Evidence Retrieval*. *Proceedings of the BioNLP 2026 (Shared Tasks)*, pages 248–256, San Diego, California, USA. Association for Computational Linguistics, 2026. DOI: 10.18653/v1/2026.bionlp-2.33
- **Basil Ebinesar**, Charansai Maddineni, Keyuan Jiang, Gordon R. Bernard. *m6A-SurvFormer: Cross-Attention over Biological Priors for Robust Survival Prediction in Lung Adenocarcinoma*. *Proceedings of the PharML Workshop, ECML-PKDD 2026*, [pages TBD]. Springer, 2026. (Accepted, spotlight presentation)
- **Basil Ebinesar**, Charansai Maddineni, Pavan Praneeth Katakam, Keyuan Jiang. *The Pride-Boiler System for the OmniRNA Discovery Challenge*. *Proceedings of the OmniRNA Discovery Challenge Workshop, ECML-PKDD 2026*, [pages TBD]. Springer LNCS, 2026. (2nd place, Naples, Italy)
- Karthikumar Sankar, ... **E. Basil Tamil Selvan** *Effectiveness of image-based classifiers for on-site identification of lichens in reserved forest*. *Computers and Electronics in Agriculture* (Elsevier), 2025. DOI: 10.1016/j.compag.2025.110994

TECHNICAL SKILLS

Deep Learning & Generative AI: PyTorch, TensorFlow, Keras, PyG (Graph Neural Networks), Transformers (Hugging Face), LangChain, LangGraph.

Bioinformatics & Scientific Computing: Scanpy, Biopython, NumPy, Pandas, Scikit-learn, Linux/Unix Shell Scripting.

Cloud & MLOps: AWS (EC2/S3 for HPC, Lambda), Docker, Git, SQL, Pinecone, FASSIS (Vector DB), SageMaker.

Languages: Python, Bash.

HONORS & AWARDS

- **Graduate Research Award:** Purdue University Northwest, 2026.
- **Gold Medalist (Valedictorian):** Awarded for securing **Rank 1** across the entire engineering cohort (2024).
- **National Finalist**, Smart India Hackathon (SIH) 2022: Selected among top teams nationwide for rapid prototyping of AI solutions.
- **Best Technical Presentation:** Awarded for "AI in Medical Imaging" research presentation at the Tech Expo Symposium.
- **Best Overall Award, App Development Challenge:** Recognized for developing *NeuroScan*, an application focused on intelligent medical data analysis.

LEADERSHIP & SERVICE

- **Senior Under Officer (Troop Commander), NCC:** Led a battalion of cadets; managed logistics for state-level camps and recognized as **Best Cadet** for discipline and crisis management.
- **Founder, Applied AI Club:** Established a student-led technical community at Purdue University Northwest, organizing workshops and Student activities on AI.
- **Invited Guest Lecturer:** Delivered technical seminars on "Cloud Architectures & AI Applications" to undergraduate engineering students.